EconKit Mathematical Notes

EconKit Project

March 21, 2026

1 Purpose

This document records the mathematical formulas implemented in EconKit and explains how each formula connects to computational economics workflows.

2 AR(1) Shock Process and Discretization

EconKit models many exogenous processes as

$$z_{t+1} = \rho z_t + \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim \mathcal{N}(0, \sigma_\varepsilon^2), \qquad |\rho| < 1. \tag{1}$$

The stationary standard deviation is

$$\sigma_z = \frac{\sigma_\varepsilon}{\sqrt{1-\rho^2}}. \tag{2}$$

2.1 Tauchen

Given a grid $\{z_j\}_{j=1}^N$, transition probabilities are

$$\Pi_{ij} = \Pr\left(z_{t+1} \in \left[z_j - \frac{\Delta}{2}, z_j + \frac{\Delta}{2}\right] \mid z_t = z_i\right), \tag{3}$$

with endpoint bins handled by one-sided tails and interior bins by CDF differences. This converts a continuous shock to a finite Markov chain for Bellman iteration.

2.2 Rouwenhorst

For persistent processes, Rouwenhorst recursively constructs Π_N with

$$p = q = \frac{1+\rho}{2} \tag{4}$$

and midpoint row scaling at each recursion. This method preserves persistence and moments better than naive coarse grids when ρ is high.

3 Quadrature and Expectations

3.1 Gaussian-Hermite for Standard Normal

For $X \sim \mathcal{N}(0,1)$,

$$\mathbb{E}[f(X)] = \int_{-\infty}^{\infty} f(x)\phi(x)\,dx \approx \sum_{k=1}^{n} w_k f(x_k), \tag{5}$$

where EconKit uses transformed Hermite nodes and weights to match normal density.

3.2 Gauss-Laguerre for Exponential Weight

For integrals with e^{-x} weight on $[0,\infty)$,

$$\int_0^{\infty} e^{-x} g(x)\,dx \approx \sum_{k=1}^{n} w_k g(x_k). \tag{6}$$

For $X \sim \text{Exp}(1)$, this directly approximates $\mathbb{E}[g(X)]$.

3.3 Discrete Expectation

For finite support,

$$\mathbb{E}[X] = \sum_{i=1}^{N} p_i x_i, \tag{7}$$

implemented as a dot product after probability validation.

4 Interpolation in Dynamic Programming

Continuation values are evaluated off-grid via linear interpolation:

$$V(x) \approx (1-t)V(x_i) + tV(x_{i+1}), \qquad t = \frac{x - x_i}{x_{i+1} - x_i}. \tag{8}$$

This is central to VFI/EGM because policy updates typically generate non-grid next states.

5 Bounded Scalar Optimization (Implemented in Session 4)

Given objective $u(c) + \beta\mathbb{E}[V(x')]$, many subproblems reduce to

$$x^* = \arg\max_{x \in [a,b]} f(x). \tag{9}$$

EconKit implements:

- Golden-section search: robust derivative-free interval contraction.
- Brent-style search: combines parabolic interpolation with safeguarded bracketing.

These methods are used when derivatives are unavailable or expensive.

5.1 Golden-Section Interval Update

Given interior points $x_1 < x_2$ in $[a, b]$,

$$x_1 = a + (1 - \varphi^{-1})(b - a), \qquad x_2 = a + \varphi^{-1}(b - a), \qquad \varphi = \frac{1 + \sqrt{5}}{2}, \tag{10}$$

the interval is contracted by comparing $f(x_1)$ and $f(x_2)$, preserving the maximizer bracket while reusing one function evaluation per iteration.

5.2 Brent-Style Step Selection

Brent-style bounded search alternates between:

1. parabolic interpolation (fast local convergence when stable), and
2. golden-section fallback (robust global progress).

The safeguarded update keeps iterates inside $[a, b]$, making it suitable for economic policy problems with hard constraints.

6 Value Function Iteration and Bellman Residual (Implemented in Session 6)

Let T denote the Bellman operator on value functions. Standard VFI iterates

$$V^{(k+1)} = T\left(V^{(k)}\right), \qquad k = 0, 1, 2, \ldots \tag{11}$$

until a fixed-point tolerance is met.

6.1 Relaxed Update

To improve stability in some models, EconKit exposes relaxation $\omega \in (0, 1]$:

$$\widetilde{V}^{(k+1)} = \omega V^{(k+1)} + (1 - \omega) V^{(k)}. \tag{12}$$

When $\omega = 1$, this reduces to plain VFI.

6.2 Bellman Residual Diagnostics

The stopping metric is the residual between successive iterates, typically the sup norm:

$$r_k^{\infty} = \left\| \widetilde{V}^{(k+1)} - V^{(k)} \right\|_{\infty} = \max_i \left| \widetilde{V}_i^{(k+1)} - V_i^{(k)} \right|. \tag{13}$$

EconKit also exposes an L^2-style residual for diagnostics:

$$r_k^2 = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \left(\widetilde{V}_i^{(k+1)} - V_i^{(k)} \right)^2}. \tag{14}$$

Convergence is declared when $r_k^{\infty} < \varepsilon$, where ε is the configured tolerance.

6.3 Howard-Style Policy Evaluation Hook (Session 7)

Given a candidate policy $\pi^{(k)}$, policy evaluation applies repeated updates under fixed policy:

$$V^{(k,j+1)} = T^{\pi^{(k)}}\left(V^{(k,j)}\right), \qquad j = 0, 1, \ldots, H-1, \tag{15}$$

where H is the configured number of Howard policy-improvement steps. This can reduce outer Bellman iterations by doing extra work per iteration when policy evaluation is cheap relative to policy updates.

7 Canonical Consumption-Savings Helper (Implemented in Session 8)

For asset state a, deterministic income y, and gross return $1 + r$, the helper solves the standard one-dimensional Bellman problem on an asset grid:

$$V(a) = \max_{a' \in \mathcal{A}} \left\{ u(c) + \beta V(a') \right\}, \tag{16}$$

subject to

$$c = (1 + r)a + y - a', \qquad c > 0. \tag{17}$$

The implementation performs brute-force search over a' on a fixed grid to produce a robust baseline policy/value solution.

7.1 CRRA Utility

The utility specification is

$$u(c) = \begin{cases} \log(c), & \sigma = 1, \\ \dfrac{c^{1-\sigma} - 1}{1 - \sigma}, & \sigma \neq 1. \end{cases} \tag{18}$$

7.2 Residual-History Summary Diagnostics

Beyond per-iteration residuals r_k, Session 8 adds summary metrics over history $\{r_0, \ldots, r_{K-1}\}$:

$$\text{avg-log-drop} = \frac{\log_{10}(\max(r_0, \epsilon)) - \log_{10}(\max(r_{K-1}, \epsilon))}{K - 1}, \tag{19}$$

with machine-safe ϵ. This statistic gives an interpretable average orders-of-magnitude reduction per iteration.

8 Stochastic-Income Canonical Helper and Persistence (Implemented in Session 9)

With income shock state z and transition matrix Π, the stochastic consumption-savings Bellman problem is

$$V(a, z) = \max_{a' \in \mathcal{A}} \left\{ u(c) + \beta \sum_{z'} \Pi(z, z') V(a', z') \right\}, \tag{20}$$

with budget constraint

$$c = (1 + r)a + y(z) - a', \qquad c > 0. \tag{21}$$

EconKit's Session 9 canonical helper uses Tauchen or Rouwenhorst transitions to compute this expected continuation-value term on a product grid.

8.1 Convergence Artifact Persistence

Session 9 adds JSON-ready persistence for convergence diagnostics. Given history $\{r_k\}$, the artifact stores:

$$\texttt{artifact} = \{\texttt{history}, \texttt{summary}, \texttt{metadata}\}, \tag{22}$$

where `summary` includes count, endpoints, monotonicity flag, and average log-drop metric. This enables reproducible solver diagnostics across runs.

9 Maintenance Rule

Update this document every session where numerical methods are added or changed. Each new method must include:

1. Mathematical statement,
2. Economic interpretation,
3. Numerical approximation details,
4. Link to implementation module.